



Sensitivity Testing of Stereophotoclinometry for the OSIRIS-REx Mission. II. Effective Observation Geometry for Digital Terrain Modeling

Eric E. Palmer¹ , John R. Weirich¹ , Robert W. Gaskell¹ , Diane Lambert², Tanner Campbell² , Kris Drozd², Olivier S. Barnouin³ , Michael G. Daly⁴ , Kenneth Getzandanner⁵ , John N. Kidd, Jr.⁶ , Coralie D. Adam⁷ , and

Dante S. Lauretta²

¹ Planetary Science Institute, Tucson, AZ 85719, USA; epalmer@psi.edu

² Lunar and Planetary Laboratory, University of Arizona, Tucson, AZ 85712, USA

³ The Johns Hopkins University Applied Physics Laboratory, Laurel, MD 20723, USA

⁴ The Centre for Research in Earth and Space Science, York University, Toronto, ON, M3J 1P3 Canada

⁵ Goddard Space Flight Center, Greenbelt, MD 20771, USA

⁶ Ascending Node Technologies, LLC., Tucson, AZ 85716, USA

⁷ KinetX Inc., Simi Valley, CA 93065, USA

Received 2023 September 6; revised 2023 December 6; accepted 2023 December 7; published 2024 February 21

Abstract

The OSIRIS-REx mission used stereophotoclinometry (SPC) to generate digital terrain models (DTMs) of its target asteroid, Bennu. Here we present a suite of preflight tests conducted to identify the observing geometry and number of images needed to create DTMs that would enable successful navigation around and to the surface of the asteroid. We demonstrate that high-quality DTMs can be generated by using only five images: four that are focused on topography, in which the spacecraft's viewing geometry brackets the target (north, south, east, and west), and a fifth that measures the target's albedo variation, taken from near local noon. We further show that the first 10 iterations of the SPC process can meaningfully improve DTM quality, including in the case of a suboptimal input image set, whereas after 10 iterations the DTM quality approaches an asymptotic maximum. We distill our findings into recommendations for observation planning that can be applied by other missions intending to use SPC to model the shape and terrain of their target.

Unified Astronomy Thesaurus concepts: Asteroid surfaces (2209); Asteroids (72); Near-Earth objects (1092)

1. Introduction

In support of the OSIRIS-REx mission to the near-Earth asteroid (101955) Bennu (Lauretta et al. 2017, 2021), the digital terrain modeling software known as stereophotoclinometry (SPC; Gaskell et al. 2008, 2023) was evaluated to ensure that it could meet the mission's demanding requirements (Palmer et al. 2024). SPC combines stereophotogrammetry and photoclinometry to generate digital terrain models (DTMs) from spacecraft imagery, the technical details of which are described in Palmer et al. (2022). Because the OSIRIS-REx spacecraft needed to fly numerous trajectories around a small (500 m diameter) body and touch down at a precise location on its rocky surface to collect a sample (Lauretta et al. 2022), accurate DTMs were essential for navigation at Bennu (Barnouin et al. 2020). A comprehensive test suite was conducted before the asteroid encounter to qualify SPC for NASA Class B flight software certification and verify its suitability for operations (Olds et al. 2015; Weirich 2022; Adam et al. 2023; Palmer et al. 2024).

During the testing of the SPC software, as described in the companion paper (Palmer et al. 2024), it became clear that the observational geometry of the input images is critical to the quality of the DTM that SPC can produce. Here we describe the extensive effort that was then undertaken to evaluate the influence on DTM quality of the number of input images, the Sun azimuth and zenith angles relative to the target,

and the spacecraft azimuth and zenith angles relative to the target. Other efforts (Barnouin et al. 2020; Craft et al. 2020; Daly et al. 2022; Ernst et al. 2023) have only summarized what factors make a first-rate SPC DTM that this work is responsible for demonstrating. We also discuss the assessment of how iterative model processing steps—specifically, the number of Align, Extract, and Solve (AES) cycles performed in the SPC software (Palmer et al. 2022, 2024)—can mitigate the effects of less-than-ideal observing geometry.

2. Simulated Data Used for Testing

The testing described here simulated DTM generation during the planned proximity operations around Bennu to identify what constitutes an effective input image set for SPC. Details of the production of synthetic data for testing SPC are described in Palmer et al. (2024), but a short summary follows.

2.1. Truth DTM

First, we generated a truth model of a synthetic asteroid, called Shape 3 (Figure 1; see Palmer et al. 2024), based on our understanding of a reasonable asteroid surface from other missions—primarily the NEAR-Shoemaker (Prockter et al. 2002) and Hayabusa (Yoshikawa et al. 2021) missions to asteroids Eros and Itokawa, respectively (Barnouin et al. 2020). Shape 3 included both topography and albedo information. It was used as the truth model for all tests except one, which used a significantly simplified model (Figure 1, as described in Section 4.3). The DTM was created with a spacing between vertices, or ground sample distance (GSD), of 5 cm globally and 1 cm over key test regions.



Original content from this work may be used under the terms of the [Creative Commons Attribution 4.0 licence](https://creativecommons.org/licenses/by/4.0/). Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

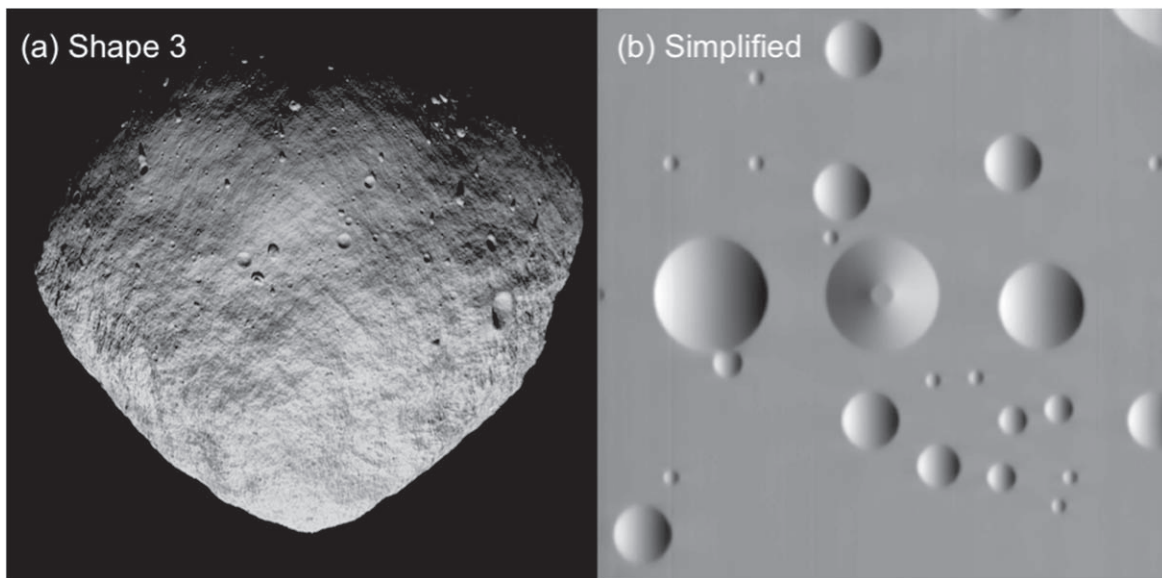


Figure 1. The two truth DTMs were used for testing. (a) The truth DTM, Shape 3, that was used for most tests. Its entire surface is rough with large boulders. These roughness attributes were generated according to observations at asteroids Eros and Itokawa (see Barnouin et al. 2020, for details). The width of the entire DTM is 500 m. (b) The simplified terrain, created to elucidate the effects of craters and hills on the performance of SPC. The slopes were well delineated, making deviations between this truth model and the SPC-derived DTM easily quantifiable. The width of simplified DTM is only 50 m.

Using the truth model, the flight profile as planned prelaunch (known as Design Reference Mission C; Lauretta et al. (2017), and the spacecraft’s expected position, we simulated observations of the asteroid. These simulated images were then fed into the SPC shape modeling process to create a DTM. Palmer et al. (2024) and Weirich (2022) describe a wide range of testing performed in this way; here we focus on simulations of the portion of the OSIRIS-REx mission known as Detailed Survey, specifically the Baseball Diamond flybys, which would be critical to global DTM generation in flight (Lauretta et al. 2017). During this phase, the plan was for the spacecraft to collect images with a pixel size of approximately 5 cm.

During testing, this original data set was shown to be inadequate (Palmer et al. 2024). Additional image sets were created that increased the number of images and the viewing geometries. These expanded image sets are described for each of the different tests.

We focused on a 50×50 m region of the Shape 3 truth model near the equator that was defined as a simulated site for the spacecraft’s Touch-and-Go (TAG) sample collection operation, known as TAG site 1. We created high-resolution SPC terrain over the entire TAG site 1 area, but we performed the evaluation on the central 20×20 m portion to ensure that the edge effects of the 50×50 m DTM did not skew the results.

2.2. Simplified Terrain

To allow isolation of some test parameters and clarity of interpretation, we created a very simple “cartoon” topography (Figure 1). Starting with a flat surface, we added mathematically generated, circularly symmetric, simple craters and peaks. The craters have a parabolic shape with a depth-to-diameter ratio of 0.2. The peaks have a 20° slope and a flat top to allow easy visual differentiation from the craters. We placed this simplified terrain near the equator on Shape 3. This DTM was only used in the F2 Test Suite.

3. Metrics for Assessing DTM Quality

To evaluate and compare the quality of DTMs produced from various imaging conditions, we used multiple metrics. Initially, the prime evaluation tool was the rms calculation of the deviation between the truth DTM and the SPC-derived test DTM (Palmer et al. 2024). In this method, each vertex of the test DTM is evaluated to find the closest location on the truth DTM. All of the distances are squared and averaged, and the square root is taken.

Another metric, which can be more informative than rms (Palmer et al. 2024), is normalized cross-correlation (Lewis 1995). This technique evaluates two images—typically a truth image and an image rendered using the SPC-derived DTM—and identifies how closely they match, where a correlation score of 1.0 means a perfect match and 0.0 means no match. During application, the rendered image is cropped and tested over the entire range of the truth image. The normalized cross-correlation test identifies the best-fit location on the truth image. This is a key technology used for flight navigation in which a DTM-derived image is compared to an actual flight image; cross-correlation of multiple features is then used to triangulate the spacecraft’s position (Adam et al. 2023). In our preflight testing, the truth image was derived from the truth model.

As the testing process advanced, a new metric called the Sigma Score (SigS) was introduced to establish a connection between correlation scores and mission requirements. For the OSIRIS-REx mission, which involved autonomous navigation to the surface, assessing the quality of the navigation features represented by small localized DTMs was crucial. There was a requirement for these localized DTMs to have a correlation score of 0.6 for the navigation software to accept the solution (Mario et al. 2022). Simply using the average correlation score to assess the quality of these localized DTMs did not provide sufficient robustness.

SigS was created to incorporate both the average correlation score and how consistent those scores were, i.e., the standard

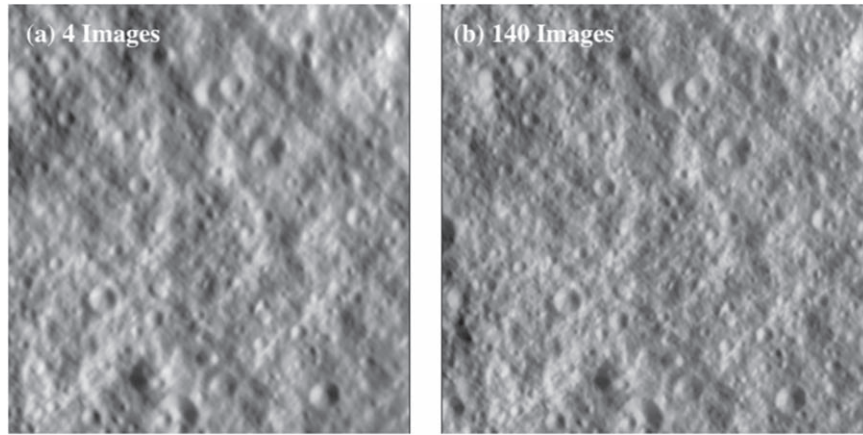


Figure 2. (a) DTM produced using four images. The rms was 7.1 cm, and the normalized cross-correlation score was 0.9373. (b) DTM produced using 140 images. The rms was 7.5 cm, only about half a centimeter better than the four-image DTM, and the normalized cross-correlation score was 0.9416. One can see that there is only a small improvement from (a) to (b), with the latter having slightly sharper edges.

deviation of the correlation scores. A good DTM would have a passing average cross-correlation score (above 0.6) and no outliers. A poor DTM could have a passing average cross-correlation score but a wide distribution of cross-correlation scores, meaning that a few images could fail while most worked exceptionally well. Additionally, a DTM with a wide distribution of correlation scores would need a higher average to ensure that the 0.6 threshold was met by most images.

Fundamentally, the SigS reports how many standard deviations the test data set performs above the mission requirement of 0.6. During testing, SigS was shown as a strong and useful metric for quality suitable for evaluation of local DTMs. It avoids a typical problem encountered by flown missions where an abundance of data from a limited range of observation geometries generates a high correlation average that masks important issues with the DTM that are only reflected in a few images. The ability to identify a sufficiently navigable DTM is essential, and images with poor correlation scores signify a weak DTM that needs improvement.

The SigS is defined in Equation (1), where $\bar{X}$ is the average correlation score and σ is the standard deviation of the evaluation images. This score quantifies the number of standard deviations by which the DTM outperforms the requirement of 0.6 assuming a Gaussian distribution. Thus, a SigS of 1 means that there is a probability of 84.1% that a rendered image will achieve a cross-correlation score of 0.6 or above. A SigS of 2 means that there is a 97.7% chance that an image will achieve a correlation score of 0.6 or above. To measure SigS, we created nine images in which to test the DTM that covered an extensive range of observation geometries, matching the test data described in Figure 5. These included an image at each cardinal direction and 45° elevation with the Sun both east and west of the location. Additionally, it included an image with a low incidence angle to capture albedo. Many tests lacked these images, so a SigS was not always available:

$$\text{SigS} = \frac{\bar{X} - 0.6}{\sigma}. \quad (1)$$

The correlation score, and especially the SigS, provided objective and responsive metrics for what constituted a quality image set. Additionally, we used the correlation and SigS scores to assess how the number of AES processing cycles (Palmer et al. 2022) influenced the DTM quality.

When the tests reference a cross-correlation score, this was done on a single image selected to be representative and covered the entire area being tested. As our testing progressed, we changed to a suite of nine images (Figure 5) to generate a SigS and the average cross-correlation score for all nine images.

The final major evaluation technique, formal uncertainty, does not use the truth model but instead evaluates how closely the DTM matches every individual piece of input data. SPC produces the formal uncertainty metric by calculating the average deviation between the center of every maplet and its corresponding position in each image. It is further explained in Palmer et al. (2024) but was used during these tests to assess its effectiveness (which can be done without a truth DTM) as a proxy for accuracy when there is not a truth DTM.

4. Tests and Results

During this study, we varied the observation geometries and number of images to identify their impact on the final SPC terrain models, as shown in Table 1. Each test built on the previous test with lessons learned, informing which imagery parameters to evaluate and how to measure model fidelity. The Design Reference Mission (DRM) and Expanded DRM are described in the companion paper (Palmer et al. 2024). The other imaging conditions are given for each test in the sections that follow. By the end of the tests, a clear set of requirements that could be applied during mission planning was developed (Section 5.1).

4.1. Number of Images (F3B)

The most frequent question posed about SPC is how many images it needs. To answer this, we studied the impact on DTM quality of varying the number of input images. This test used the 50 × 50 m area called TAG site 1 on Shape 3. This test included seven unique images that covered the entire test region. These seven images were duplicated 20 times to make additional images that were given different navigation errors. Uncorrected navigation errors can cause significant problems in a DTM, so this ensures that SPC spacecraft update procedures are effective in a realistic environment. Each image had a slightly different spacecraft position and pointing, adjusted to be within the 1σ actual navigational solution (i.e., each cloned

Table 1
Sensitivity Tests

Test Name	Data Set ^a	DTM	Identifier
Number of images	Expanded DRM	Shape 3	F3B
Expanded observation conditions	Non-Bennu-like ^b	Shape 3	F2: Test 5 & 8
Expanded flight observation conditions	Custom	Simplified terrain	F2: Test 11
Minimum flight observation conditions	Expanded DRM	Shape 3	F3C
Limited imagery	Expanded DRM	Shape 3	F3C-Systematic
Amount of processing	DRM	Simplified terrain	F2: Test 3

Notes.^a The DRM and Expanded DRM image suite is detailed in the companion paper (Palmer et al. 2024).^b The non-Bennu-like data set was a suite of ideal data used to evaluate the ideal performance of SPC. This data set included solar geometries that cannot exist on Bennu owing to its obliquity, although they can exist on other bodies.**Table 2**
Quality of DTMs Produced from Varying Numbers of Images (F3B)

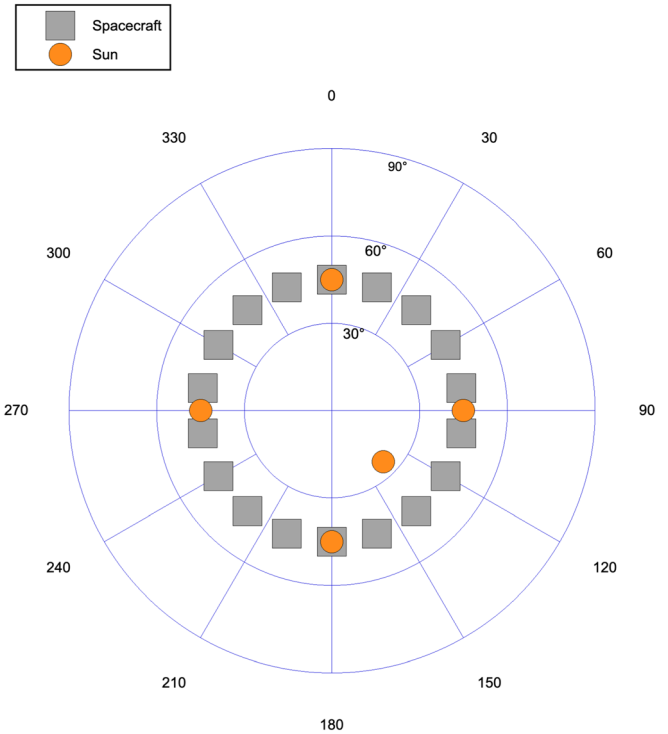
Number of Images	rms (cm)	Correlation Score
4	7.7	0.9373
7	7.1	0.9339
14	7.1	0.9434
70	7.5	0.9409
140	7.5	0.9416

image had a different displacement). We did not create new viewing geometries for this early test, so the impact of having excess duplicate data with different positional errors is evaluated.

We grouped the images into five sets, each with a different number of images: 4, 7, 14, 70, and 140. We created a DTM of the test region from each image set using standard SPC techniques (Palmer et al. 2022). We conducted 15 AES iterations for each test. Each of the resulting DTMs was compared to the truth model (Shape 3, the original source of the input images) to compare the effectiveness of the different image sets. The rms errors between the truth and test DTMs (Table 2) are relatively consistent, ranging from 7.7 to 7.1 cm—that is, less than 1 cm of variability between all of the different image sets. These results suggest that SPC only requires a limited number of images to reach a stable solution—i.e., four images at an improved pixel size can provide a qualitative improvement of the low-quality starting DTM.

The correlation score also showed only a minimal difference between each data set, from a minimum of 0.9339 to a maximum of 0.9434, or less than 0.01 difference (Table 2). A visual analysis of the rendered images indicated only a trivial difference. If the output from the four-image test is compared to the 140-image test, one can note that some features are slightly sharper; see Figure 2. This is likely because every image is processed during each AES iteration, which means that more data are being pushed into the DTM even though the number of iterations is the same.

Nevertheless, there is some variation among both the rms and correlation scores. We assess most of this to be within the noise created by having different numbers of images and different spacecraft position errors. The tests described in the following sections have larger variation in the results and provide a better indication that a variety of viewing geometries is more critical than the number of images.

**Figure 3.** Imaging geometries for the Expanded Observational Conditions test. The polar plot shows the position of the spacecraft (gray filled squares) and the position of the Sun (orange filled circles). Both plots are viewed from the surface, looking up. The radial gridlines indicate azimuth, where north is 0° and east is 90°. The concentric gridlines indicate the zenith angle, with 0° being straight up.**4.2. Expanded Observational Conditions Test (F2: Tests 5 and 8)**

The next test studied how much the SPC accuracy could be improved by increasing the number of camera orientations from the original OSIRIS-REx mission plan. Using the data from Shape 3, additional observing stations were generated for the TAG 1 site so that there were viewing stations at 45° of elevation at every 20° of azimuth.

We used polar plots to describe the observation stations' geometry (Figure 3). A polar plot indicates the position of the Sun and spacecraft relative to a specific region on the surface. This position is determined using incidence (Sun to surface normal) and emission (spacecraft to surface normal) vectors. The angular component is the azimuth of either the Sun or

Table 3
Performance of SPC at Each DTM Processing Step (F2: Tests 5 and 8)

Resolution	Processing Step	Image Pixel Size (cm)	rms (cm)	Formal Uncertainty (cm)	Correlation Score
Standard	35 cm GSD	5 cm	8.58	5.72	0.6423
Standard	18 cm GSD	5 cm	6.58	4.22	0.7381
Standard	9 cm GSD	5 cm	4.92	2.26	0.8113
Standard	5 cm GSD	5 cm	4.45	1.35	...
High ^a	2.0 cm GSD	0.5 cm	1.28	1.35	0.8507
High ^a	1.00 cm GSD	0.5 cm	1.19	1.34	0.8953
High ^a	0.50 cm GSD	0.5 cm	1.18	1.27	0.9048

Note.

^a The high-resolution tests were added for the evaluation of the Natural Feature Tracking autonomous navigation system (Olds 2014). The testing location was a 2×2 m subsection of the TAG site; being for a significantly smaller area, the results should be considered a different location.

spacecraft; north (0°) is up. The radial component defines the number of degrees from the zenith point (also defined as 90° minus elevation). Figure 3 shows where the spacecraft and Sun are in the sky relative to a specific point on the surface. Describing the observation stations using only incidence and emission angles is insufficient because those parameters collapse a vector into just magnitude.

The illumination for this test included observations with the Sun in the north, south, east, and west with a 45° zenith angle and one image with the Sun closer to local noon, specifically in the southwest at 30° zenith angle (Figure 3). Previous testing of SPC suggested that this image set would be one of the best available. These conditions cannot physically occur on Bennu because Bennu has an obliquity of near 0° (Nolan et al. 2013), i.e., the Sun will not be in the north or south; however, we wanted to determine the performance during the best possible case. Hence, images were included that could only exist if Bennu had a significantly higher obliquity and the data were collected over an entire Bennu year.

This test comprised two sections: a “standard-resolution” section with an image pixel size comparable to the OSIRIS-REx mission prime requirements (5 cm) and a “high-resolution” section with an image pixel size of 0.5 cm to evaluate SPC’s ability to support additional navigation responsibilities (Olds et al. 2015). The image set consists of a series of images with pixel sizes of 50, 20, 10, and 5 cm for the standard-resolution part, with images with pixel sizes of 1 and 0.5 cm added for the high-resolution portion. Table 3 shows evaluation metrics that were captured during the processing at various stages of the DTM generation. Iterative processing at decreasing GSDs is described in Palmer et al. (2022). The images and DTMs are processed the same way as the mission simulation tests in Table 3 of Palmer et al. (2024).

This test confirmed expectations: as the GSD of the maplets and resolution of the imagery increased, the quality of the DTMs improved. The rms error was closely related to the image pixel size, being 4.45 cm at the end of the standard-resolution section and 0.84 cm at the end of the high-resolution section. The formal uncertainty was also tracked with the processing steps (the generation of smaller and smaller maplets; Palmer et al. 2022) and was limited by the image resolution. The standard-resolution section, at 5 cm GSD, reached an impressive 1.35 cm and 1.27 formal uncertainty, respectively. Figure 4 demonstrates the visual impact of higher-resolution images and higher-GSD maplets.

4.3. Expanded Flight Observational Conditions, Simplified Terrain (F2: Test 11)

The test in Section 4.2 provided a wealth of information about the performance of SPC, especially with regard to the impact of image resolution; however, the entire system—SPC procedures, Sun position, spacecraft position, terrain ruggedness, and quality of images—contained a large number of parameters, making it difficult to isolate which had the greatest impact.

This next suite of tests used the simplified terrain (Figure 1(b)) to remove the effect of surface roughness on SPC performance and isolate that of observational geometry. This test involved three major input image sets that have variations in the following: spacecraft azimuth angle, spacecraft elevation angle, and solar elevation angle (Table 4). Unlike the prior test, we restricted the Sun’s positions to those that are physically possible on Bennu, that is, a subsolar latitude near the equator. The first set of subtests varied azimuth distributions of spacecraft angles from 20° to 90° (subtests 11F, 11G, 11H, 11I, and 11J). The second set varied the number of unique solar positions (subtests 11K, 11L, 11M, 11N, and 11P) and included exploring the effects of single versus multiple solar positions with the same incidence angles, as well as a wide array of solar positions. The third set varied the zenith angle of the spacecraft from a small number of angles to a larger number of angles (subtests 11S, 11T, and 11U).

There were sufficient data to generate high-quality DTMs in all cases. Test DTMs had an average rms error of 0.76 cm, which is a factor of 6 better than the source image pixel size. These tests showed that SPC can operate effectively with far fewer images (4–10 producing good results depending on observing geometry) than the number of images typically used (50–200). It is also likely that the simplified terrain, with its extensive smooth regions, allowed the rms error to be smaller than typically seen with more rough terrain. The formal uncertainty and correlation scores were not recorded for the higher-resolution tests because they did not have requirements ascribed to them.

4.4. Minimum Flight Observing Conditions (F3C)

The previous tests evaluated the performance of SPC when there were more than sufficient data to generate a DTM. This next test limited the number of observing stations to determine the minimum number of images that are needed before the quality of a DTM changes significantly. For this purpose, 13

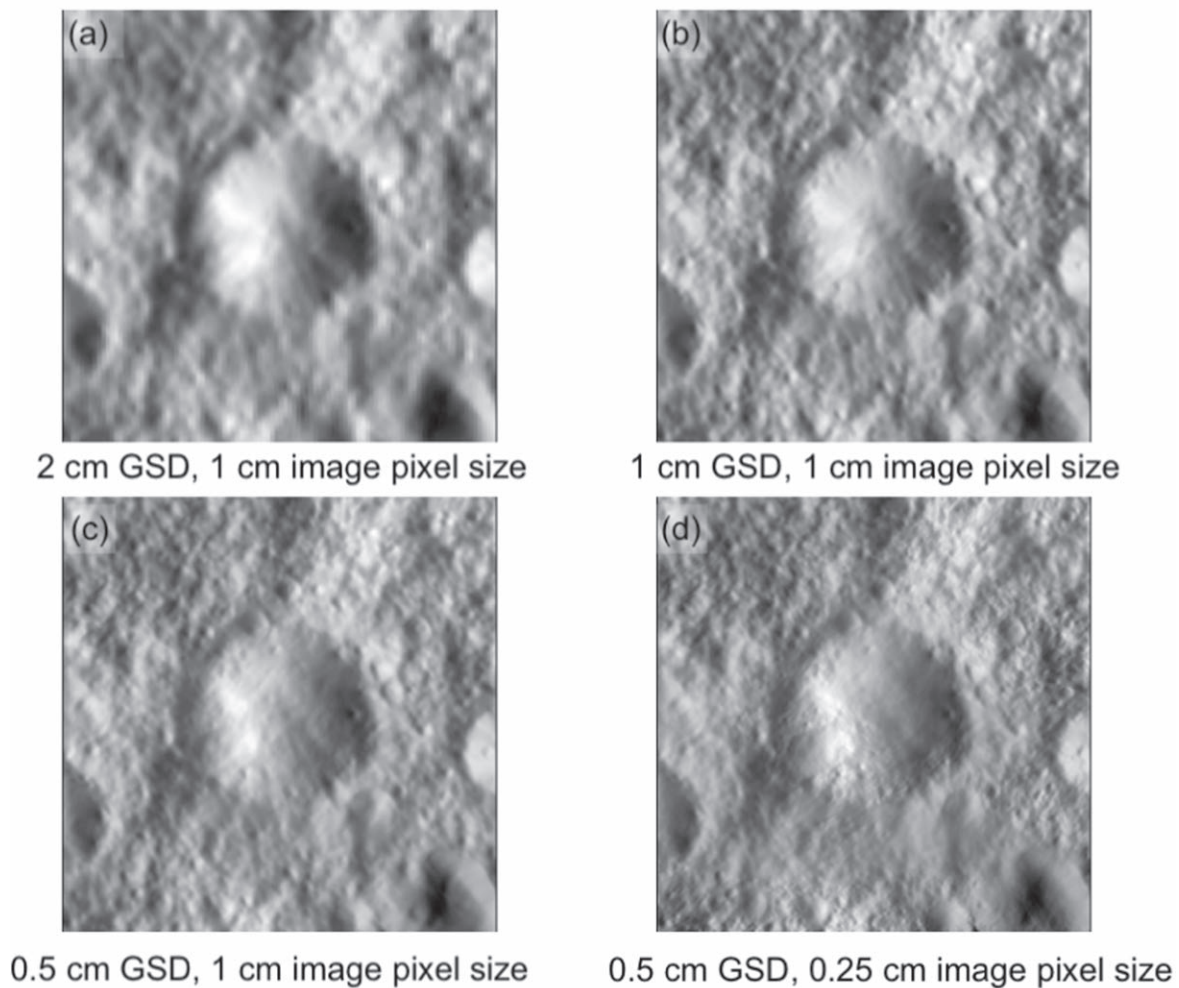


Figure 4. The results of input data and processing on the DTM quality. Each image was rendered with a 1 cm pixel size and only differed in the amount of processing applied. The DTMs represented by panels (a), (b), and (c) were generated using the same imagery but had different maplet GSDs. One can see the improvement in DTM quality due to better image data. Panels (c) and (d) show the improvement in DTM quality with higher-resolution images even if the maplet GSD does not change. The quality of terrain that SPC can generate is a function of both the pixel size of the imagery and the GSD of the desired DTM.

subtests were undertaken, each considering a different quantity of images and observing geometries (Table 5).

Similar to the previous tests, synthetic images were generated from Shape 3 with custom observation geometries and a 5 cm image pixel size. The spacecraft position was 45° zenith, and an image was simulated every 45° azimuth. The Sun position was 45° zenith, with half of the images with the Sun in the east and the other half with the Sun in the west. From this core suite of images, we created different combinations of conditions. Some variations in conditions, mostly regarding the zenith angle of the Sun, were not investigated. Although experience suggests that the zenith angle of the Sun does not strongly impact the solution, this has not been rigorously tested, which remains an opportunity for future work.

These tests show the need for the Sun, in at least some of the images, to be on the opposite side of the imaged region from the spacecraft (cross pairs). When the spacecraft and the Sun are on the same side of the target (matched pairs; e.g., a low phase angle), the solution is adversely impacted. Specifically, in the first subtest (F3C-1), which used four images that were matched pairs, we were unable to generate a DTM; during the iterative process (Palmer et al. 2022), the solution did not

converge. However, by putting the Sun on the opposite side of the terrain (subtest F3C-2), we are able to generate a DTM of moderate quality.

It was possible to generate a DTM from three images, but the quality was significantly reduced by the lack of data (F3C-3 and F3C-4). The positions of the Sun and spacecraft had more variations than in F3C-1, which is probably why these DTMs converged. Although the rms quantification of accuracy is not much lower, the correlation score is considerably lower (below the 0.6 requirement), indicating that the DTM is poor and might not be a reasonable representation of the surface.

Subtests F3C-X0 to F3C-X6 showed that having an image with a low incidence angle is important but having one with a low phase angle is not. Low-incidence images, which we call albedo images, have the Sun in a position where the impact of sloping surfaces (i.e., topography) is minimized with respect to albedo variation. This provides a discrete image type with unique data, making it easy to solve for albedo in the SPC equations.

From this suite of tests, we identified a core set of five observation geometries that provide excellent results (Figure 5). Four of the five images are optimized for modeling topography; they have respective spacecraft azimuths of north,

Table 4

Performance of SPC at Different Spacecraft and Sun Positions (F2: Test 11)

Subtest ID	Sun Azimuth (deg)	Spacecraft Azimuth (deg)	Spacecraft Zenith (deg)	rms (cm)
Spacecraft Azimuth				
11F	45, 0, -45	every 20	45	0.43
11G	45, 0, -45	every 30	45	0.44
11H	45, 0, -45	every 45	45	0.52
11I	45, 0, -45	every 60	45	0.53
11J	45, 0, -45	every 90	45	0.65
Sun Variations				
11K	30	every 45	45	1.64
11L	30, -30	every 45	45	0.87
11M	45, -45	every 45	45	1.05
11N	30, 0, -30	every 45	45	0.68
11P	45, 30, 0, -30, -45	every 45	45	0.96
11Q	60, 45, 30, 0, -30, -45, -60	every 45	45	0.96
11R	60, 30, -30, -60	every 45	45	1.16
Spacecraft Zenith				
11S	45, 0, -45	every 45	0, 30, 60	0.53
11T	45, 0, -45	every 45	0, 20, 40, 60, 80	0.52
11U	45, 0, -45	every 45	0, 14, 30, 45, 60, 75	0.55

east, south, and west, with the Sun positioned at 45° zenith angle. These geometries provide high-quality stereo angles and medium Sun angles.

The fifth image is an albedo image, in which the Sun is near local noon. When the incidence angle approaches zero, the shading due to topography becomes negligible, and the albedo signature is the strongest, thus providing a strong data set for disambiguation between the albedo and slope in the photoclinometric solution. For latitudes above or below the subsolar latitude, the topographic contribution is not entirely removed because the incidence angle cannot reach 0° , but the contribution of the topography is nevertheless at its lowest.

We used these criteria to define the imaging conditions throughout the OSIRIS-REx mission (Barnouin et al. 2020), and they were key to evaluating the quality of SPC products in flight (Al Asad et al. 2021).

4.5. Systematic Evaluation of Limited Imagery (F3C-Systematic)

With a clear idea of a reasonable suite of images that results in a complete and detailed DTM, we investigated the specific contribution of each observation geometry (Table 6). The strategy for testing this was to remove images from the input data set and note the change in performance systematically. These tests started with a DTM generated from simulated Preliminary Survey observations (Lauretta et al. 2017)—two albedo images (A and B) and four topographic images (north, east, south, and west)—and ran for at least 50 iterations of the AES process in the SPC software (Palmer et al. 2022; see

Figure 6). We evaluated the effects of removing one or both albedo images and each of the topographic images in turn (Table 6 and Figure 8).

These tests were built using images from the original mission profile for Approach and Preliminary Survey. With SPC, any images that are within a factor of four in resolution are typically used in the processing unless specifically excluded. As such, because the synthetic Preliminary Survey images have a pixel size of 15 cm, two images were included in the testing data set. Having a progression of low-to-high-resolution images is a typical technique for SPC because it provides a more stable and self-consistent solution.

In all cases, we were able to generate terrain that eventually reached a high average correlation score near 0.8 (Figure 8). The SigS provides a clearer metric for quality; we established a threshold of $\text{SigS} = 2.0$ to indicate when a data set performs at an acceptable level to meet mission requirements. After 10 iterations of the AES procedure, three data sets did not achieve the required SigS of 2.0, indicating that these are weak data sets. By continuing to process the same data via additional AES iterations, we increased the SigS such that the scores ranged from 2.14 to 4.4 after 50 iterations. However, this improvement came at a large expenditure of processing time since each AES cycle is time-consuming, and some data sets (those where one albedo image and the south topographic image were removed) only modestly exceeded a SigS of 2.0 after this effort.

When evaluating the improvement of the SigS based on the number of AES iterations for the strongest data sets, it showed that after four AES iterations approximately 50% of the maximum SigS was reached. After 10 AES iterations, 66% of the maximum SigS was reached. Beyond 10 AES iterations, each following iteration's improvement was less than 2% of improvement. Beyond 50 AES iterations, each AES iteration improved the SigS by less than 0.2%. For weaker data sets, it takes even more AES iterations to achieve the same fraction of that data set's maximum SigS.

We observed a difference in the impact of removing north versus south images. Theoretically, one would not expect much difference between these two cases. However, the data set used for this evaluation included two of the original images from the Preliminary Survey flight profile, both of which were in the north. These lower-resolution images provided a basis for the alignment of the higher-resolution images (5 cm pixel^{-1}), thereby increasing the quality of the model and reducing the impact of the loss of the northern images.

These two additional Preliminary Survey images provide stability for generating the DTM even though they are lower in resolution. The position of these images is north-northwest and north-northeast of the test site. Thus, the removal of the north topographic image impacts the quality of the solution less than the removal of the south topographic image. This shows that useful topography can be generated from images that are extrapolated by a factor of at least three. Other testing has shown that good data can be extrapolated even to a factor of four (Palmer et al. 2016). When we remove the southern topographic image, it takes about 33 iterations of the AES process to reach a SigS of 2.0—an extreme amount of computer processing time—and the SigS after 50 iterations only reaches 2.14 or 2.52 (Table 6).

Two major trends are noted. First, as more iterations of the AES process are conducted, the DTM improves; however, there is a limit to the improvement possible, regardless of the

Table 5
Performance of SPC with Minimum Flight Observing Conditions (F3C)

Subtest ID	Images	Sun Azimuth (deg)	Sun Zenith (deg)	Spacecraft Azimuth (deg)	Spacecraft Zenith (deg)	rms (cm)	Corr. Score	Sigma Score
Spacecraft Azimuth								
F3C-1 matched pairs ^a	4	90, 270	45	90, 270	45	Fail	Fail	Fail
F3C-2 cross pairs ^b	4	270, 90	45	90, 270	45	9.89	0.687	0.74
F3C-3 matched pairs ^{a,c}	3	260, 270	45	260, 270	45	11.07	0.652	0.74
F3C-4 cross pairs ^{b,e}	3	90, 100	45	260, 270 only	45	10.62	0.574	-0.17
F3C-5 single side	8	270 only	45	every 45	45	9.90	0.712	0.95
F3C-6 both sides	8	90, 270	45	every 45	45	9.25	0.770	1.75
F3C-6+ add low inc ^c	10	45, 270	10, 45	every 45	10, 45	...	0.802	2.27
F3C-X0 partial cardinal	4	90, 270	45	N, NW, S, SW	45	...	0.717	0.95
F3C-X1 cardinal	4	90, 270	45	N, E, S, W	45	...	0.738	1.14
F3C-X2 add low inc ^c	6	90, 270	2 × 10, 4 × 45	2 × N, E, 2 × S, W	4 × 45 and 2 × 10	...	0.788	2.05
F3C-X3 add low phase ^d	5	90, 270	1 × 10, 4 × 45	N, E, S, 2 × W	4 × 45 and 1 × 10	...	0.740	1.19
F3C-X4 partial low phase ^f	4	90, 270	2 × 10, 2 × 45	2 × E, 2 × W	2 × 45 and 2 × 10	...	0.732	1.26
F3C-X5 partial low inc ^g	4	90, 270	2 × 10, 2 × 45	N, E, S, W	2 × 45 and 2 × 10	...	0.734	1.24
F3C-X6 add low inc ^c	5	90, 270	1 × 10, 4 × 45	N, E, 2 × S, W	4 × 45 and 1 × 10	...	0.804	2.04

Notes.

^a Matched pairs: the Sun and the spacecraft were on the same side of the target.

^b Cross pairs: The Sun was on a different side than the spacecraft.

^c Add low inc: added an image with the Sun at a low incidence angle.

^d Add low phase: added an image with a low phase angle.

^e These tests had a small offset in the azimuth angle of 10°, which was enough to generate a DTM.

^f Removed the north/south images from F3C-X1 and replaced with low phase.

^g Change solar incidence angle for two images to low 10° from F3C-X1.

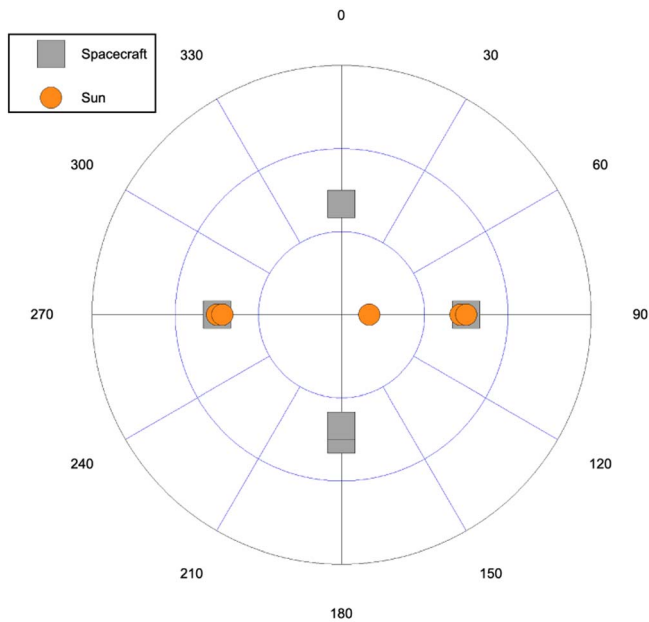


Figure 5. Polar plot of the positions of the simulated images as viewed from the surface. The positions of the spacecraft are gray squares, and the Sun is the orange circle. This was a result of test F3C-X6, in which we identified a suite of images that provided a good data set for SPC. Topography is determined by the spacecraft at cardinal directions, with the Sun on each side of the location. Albedo is determined by the Sun at a low incidence angle and, in this case, with the spacecraft south of the location.

amount of processing. The quality of the DTM, as measured by both the normalized cross-correlation score and SigS, quickly increases over the first 10 AES iterations but then improves only asymptotically approaching a maximum (Figures 6 and 7).

Table 6
Effects of Removing Individual Images from the Minimum Image Set

Subtest ID	Albedo Removed	Topo Removed	Correlation Score, 50 AES Cycles ^a	SigS, 10 AES Cycles	SigS, 50 AES Cycles
Base	...	...	0.845	2.97	4.44
Toss1	A	...	0.835	2.45	3.75
Toss2	A & B	...	0.817	1.79	2.72
Toss1S	A	South	0.810	1.79	2.52
Toss1E	A	East	0.825	2.59	4.00
Toss1N	A	North	0.822	2.02	3.32
Toss1W	A	West	0.835	2.56	4.41
Other1	B	...	0.835	2.40	3.82
Other1S	B	South	0.798	1.56	2.14
Other1E	B	East	0.831	2.76	4.32
Other1N	B	North	0.831	2.04	3.90
Other1W	B	West	0.833	2.39	4.30

Notes. The imagery set started with two albedo images (A and B) and four topographic images (one from each cardinal direction). The table shows which images have been removed.

^a Some tests were run longer than 50 cycles, so their maximum correlation score is a bit higher than shown in this table; however, by reporting the scores at 50 iterations, we ensure that the data are being evaluated consistently.

Second, the higher the quality of the input data, the faster the DTM converges. In Figure 6, significant improvements are evident during the first 10 iterations. After that, improvement continues, but the benefit of the increased processing time diminishes. When generating very high-resolution DTMs, an AES cycle can take weeks to months depending on the number of maplets, number of images, and capabilities of the

computers being used. Operational constraints may impact how much computer time can be given to refine a DTM, which can be made shorter with higher-quality observation stations.

Unexpectedly, there was only a modest decrease in performance between using two albedo images and using only one (maximum SigS of 4.4 and 3.75, respectively, a decrease of 14%; Table 6, Figure 8). However, not having an albedo image at all was a significant problem for the quality of the DTM. The decrease of the SigS from two albedo images to no albedo images was 38%, and that from one albedo image to no albedo images was 27%. After 10 AES iterations, an image set with no albedo images yields a SigS of only 1.79, less than the SigS of 2.0 established as a requirement. Thus, for OSIRIS-REx’s exceptionally high operational requirements, one albedo image is necessary, but two are not.

4.6. Amount of Processing

During SPC processing, the quality of the terrain improves with more integration time—i.e., iterations of the AES process. The first few iterations provide a large improvement in the topographic and albedo solutions, as indicated by the correlation and SigS scores; then, as the processing continues, the improvement of the model slows down, and the scores approach a maximum with diminishing improvement after every iteration (Figure 6).

For a robust and high-quality data set, such as the “Base” set from Table 6, four iterations of the AES process were sufficient to achieve a SigS of 2.0, which meets mission requirements (Figure 7). This is the absolute minimum number of iterations that we recommend for the execution of the AES process. A substantial improvement in SigS can be achieved in the first 10 iterations of the AES process, but beyond that, the improvement begins to level off. Most of the time, the data volume is very large and the processing time becomes a limitation. We therefore recommend a standard of 10 iterations of the AES process to be completed before a DTM is released for common use.

For a lower-quality data set, additional processing may be worthwhile. Section 4.4 shows that although a lower-quality suite of images will not reach the same level of fidelity that a high-quality image set will achieve (Table 6), it can still produce a DTM of usable quality, with additional processing. Between 15 and 32 AES iterations with a low-quality data set can result in a SigS of 2 (Figure 8).

Figure 9 shows the rate at which the iterative processes of SPC adjust the topography from when the model is first generated. In this example, the DTM starts as a flat surface, with each AES process increasing slope as the image’s Digital Number (DN, i.e., pixel value) is translated into the topography. It takes about 10 iterations of the AES process to reach about 80% of the crater depth. If more processing can be performed, improvements are made until about 25 AES iterations, after which the topographic fidelity closely approximates the asymptotic value.

5. Discussion

5.1. Quality of Data

In order to understand what constitutes a high-quality image set, the imaging parameters are broken into different components that allow us to characterize the influence of each on the final DTM. Although these parameters overlap with one

another, this approach allows the creation of a suite of requirements against which a mission’s imaging campaign can be planned. Specific recommendations for imaging requirements are provided in Appendix A.1; these were the imaging requirements ultimately used by OSIRIS-REx in flight (Lauretta et al. 2021).

SPC can be thought of as having two major components: the stereo component and the photoclinometry component (Gaskell et al. 2023). The stereo component provides the position of the center of each maplet in 3D space. The photoclinometry component provides the pixel-to-pixel high-resolution terrain representation. The photoclinometry component can be subdivided into a slope subcomponent and an albedo subcomponent. Because we solve for both the slope and albedo, it allows SPC to use multi-image, 2D photoclinometry to avoid many of the issues of traditional photoclinometry (Kirk 1987; Palmer et al. 2016). We discuss each component separately to allow the assessment of an optimum image suite for each component.

5.1.1. Stereo Component

The stereo component requires two images that fix the position of a feature in 3D space. By using triangulation (when the position of the spacecraft is known), the pointing, and the pixel/line of a feature in each image, we calculate the feature’s position in 3D space. The critical component of this calculation is the angle between the two observing stations; 90° is the optimum angle for stereo, so images with a 45° emission angle on either side produce the smallest amount of error. The angle can be decreased to 60° (30° emission angle) while only increasing the typical error by $\frac{1}{2}$ of an image pixel. Thus, the range of angles that produce acceptable results is very wide: emission angles between 15° and 65° can be effective as long as they are on opposite sides of the feature. A simple rotation movie (a series of images that cover the entire rotation of the asteroid every 30° or less) captured during the approach meets this requirement easily. Because the angle range is large, navigational errors can be minimized since the position of the feature can be determined from these two orthogonal vectors.

5.1.2. Photoclinometric Component

The photoclinometric component is the pixel-to-pixel height derived from multi-image 2D photoclinometry. SPC uses photoclinometry to calculate the slope for every pixel in a maplet using the data from every pixel in every image assigned to that maplet. Once all of the slopes are calculated, the slopes are integrated to form a height for every pixel. Traditional photoclinometry is based on a 1D analysis of a single image that has a uniform albedo, and the slope is calculated along the steepest direction (Kirk 1987). Any nonconformity to these constraints results in systematic errors. Additionally, any image artifacts will be folded into the DTM because the solution is not overconstrained.

SPC expands on traditional photoclinometry by solving for the x -slope, y -slope, and geometric albedo—multi-image 2D photoclinometry (Palmer et al. 2016). In this way, a full 3D model, including albedo, can be constructed, allowing the terrain to be modeled given any Sun or spacecraft position. Once this is available, a synthetic image can be rendered from the DTM with the same observing geometries as any actual image. If the DTM is a good representation, the rendered image will be nearly identical to the image taken by the spacecraft. Any

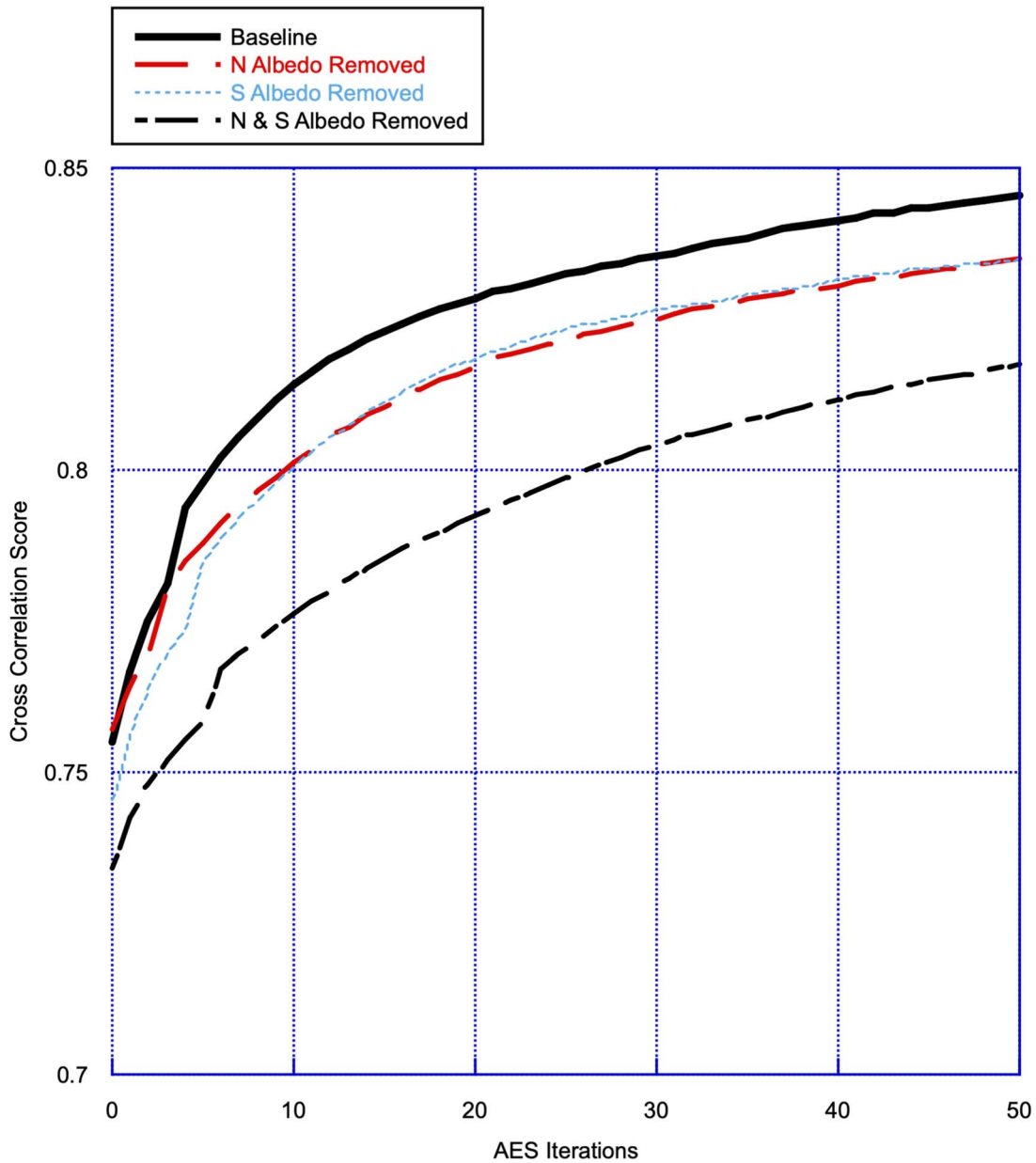


Figure 6. The average correlation score for DTMs generated with different numbers of albedo images: the full six-image input set with two albedo and four topography images, a five-image set with the Albedo A image removed, a five-image set with the Albedo B image removed, and a four-image set with neither albedo image. The x-axis is the number of iterations of the AES cycle that were run. The y-axis is the normalized cross-correlation score. By this metric, the DTM improves quickly at first, with the first 10 AES iterations accounting for most of the improvement. AES iterations beyond 50 result in only a small improvement in the DTM.

differences between the spacecraft image and the rendered image can be used as a correction for the DTM (Palmer et al. 2022). Not only does this decrease errors caused by artifacts (cosmic rays, uncorrected distortion, etc.), but it also allows for a very high-resolution DTM. This technique was used for optical navigation of the OSIRIS-REx mission (Norman et al. 2022; Olds et al. 2022; Adam et al. 2023).

To characterize an effective imagery set for 2D multi-image photoclinoetry, we evaluated the respective specifications of topographic and albedo images (Figure 10).

Topographic Images.—Topographic images are images where the variations in the pixel DN values are mostly due to the slopes within the terrain. This occurs when the Sun is low on the horizon. There is no exact requirement on the solar zenith angle, but slopes on the surface dominate the DN

variations of the topography when the zenith angle is 30° or more. Generally, SPC attempts to get the solar zenith angle to be 45° but no more than 70° .

To have a robust topographic suite of images, there are two general requirements. First, there must be images with the Sun on opposite sides of the feature. This ensures that what is shadowed in one image is not shadowed in the other. Second, observational geometries must support emission angles oriented along both horizontal orthogonal directions. Complementary images on opposite sides create a focus for both sides of any slope. Thus, one would want four images, two pairs oriented on opposite sides in two orthogonal directions, such as cardinal angles of north, east, south, and west, or they can be rotated (e.g., 45° , 135° , 225° , 315° would also be acceptable).

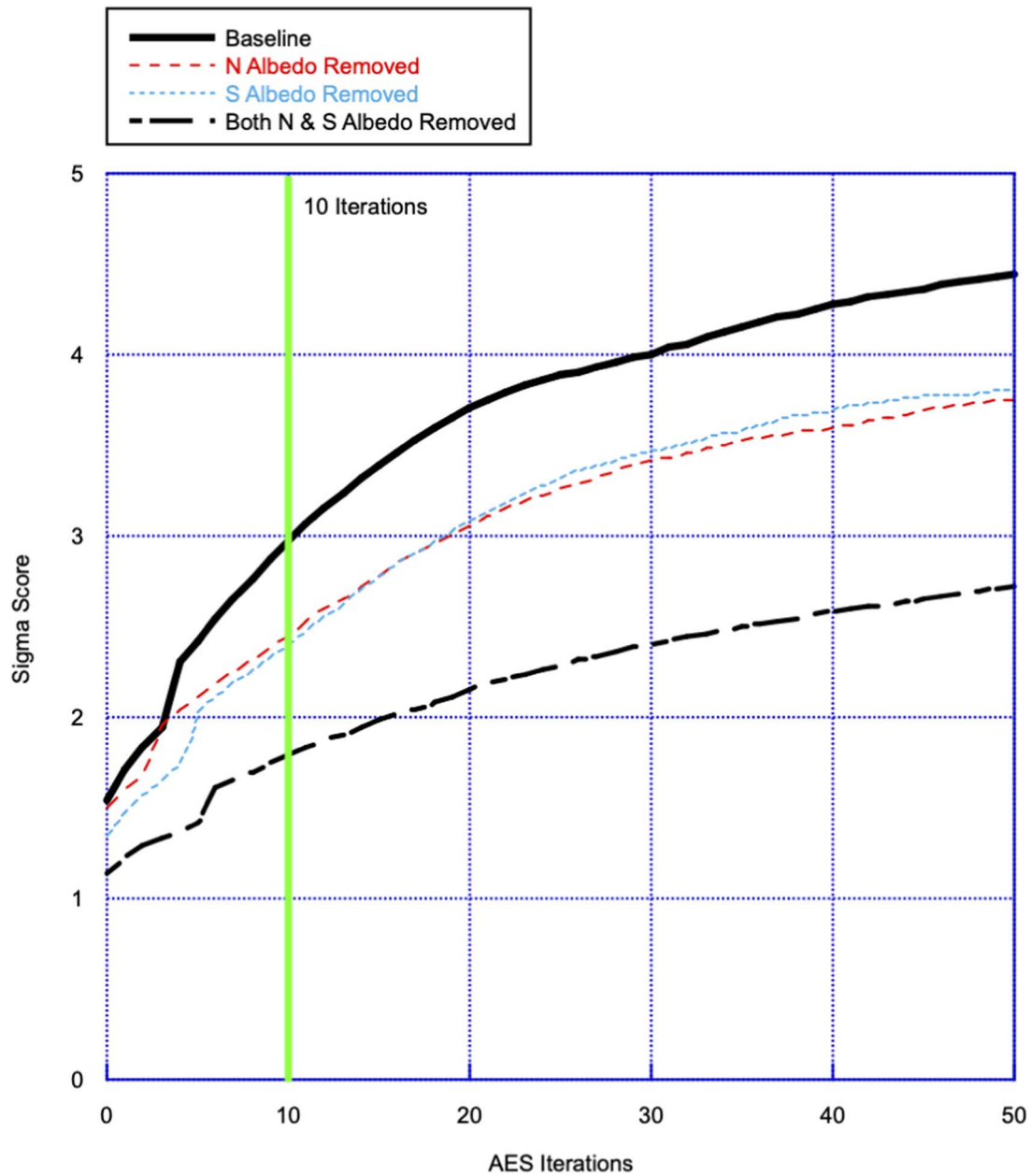


Figure 7. Same as Figure 6, but for the sigma score, SigS.

Albedo Images.—An accurate representation of the surface brightness, or albedo, is key to rendering realistic images, identifying features, and using multiple images in DTM generation.

For any image where the Sun is directly overhead (incidence angle = 0°), almost all of the topographic signature will be suppressed, and variations in the images' DN will be exclusively driven by albedo. An image set that includes an image of this kind provides the strongest solution for the albedo portion of the SPC equations. Photometric effects must be removed, but that can easily be done via the photometric function that is embedded in the SPC code (Gaskell et al. 2008).

For a practical definition of albedo images, we suggest that the Sun's zenith angle be less than 20° . Care must be exercised to avoid a phase angle that is lower than 7° because the direct backscatter of light is difficult to remove from the image data.

As such, SPC explicitly suppresses data with a phase angle less than 7° . While albedo can be extracted from any image, the higher expression of albedo seen at low zenith angles allows for faster convergence of the data and higher fidelity of the DTM solution.

5.2. Flexibility of Imaging

In general, SPC can use any image of an object as input. SPC is able to account for the exact observing geometry of every image, which means that it makes photometric corrections to the images due to the slope of the surface (i.e., the photometric function takes into consideration the incidence, emission, and phase angles at the pixel level). SPC uses three different observing geometries to form three equations that solve for the three unknowns: two orthogonal slopes and albedo (Gaskell et al. 2008, 2023; Palmer et al. 2016, 2022). Although a

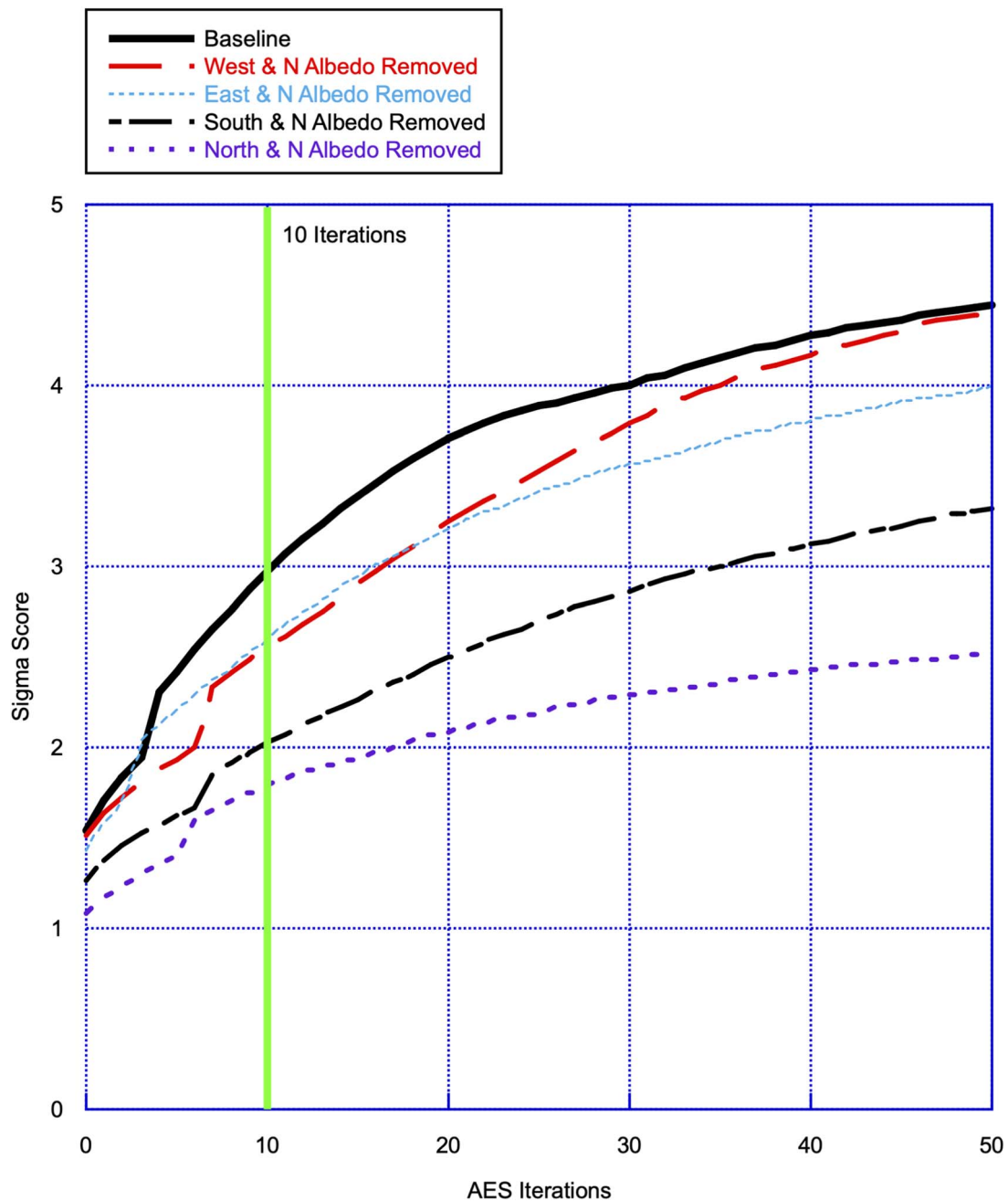


Figure 8. Same as Figure 7, but for the removal of specific topographic images.

minimum of three images are needed, in practice each piece of terrain is usually captured in many more—typically over 50.

While we present the testing and identification of a suite of images that allows for a high-quality DTM, good DTMs can be generated with observation parameters that do not meet the requirements described in Section 4.5. Many planetary missions have used SPC to model terrain by collecting whatever images the science team requested, which typically do not match optimal observing conditions for SPC yet still generate a usable DTM.

6. Conclusion

We have characterized several aspects of the SPC technique’s performance; key takeaways:

1. Observing conditions are more important than the number of images.
2. A good set of imagery has both albedo and topography-focused images.
3. 10 processing iterations provide a DTM with approximately 66% of its maximum SigS.

SPC has generated high-quality data using various observation conditions for many missions (Palmer 2022). Our testing here allows the determination of the sensitivity of the SPC technique to the source images regarding the quality of the DTM.

First, the sheer number of images is not the most essential variable in the quality of the DTM. Other factors, specifically a variety of observing conditions, are more critical.

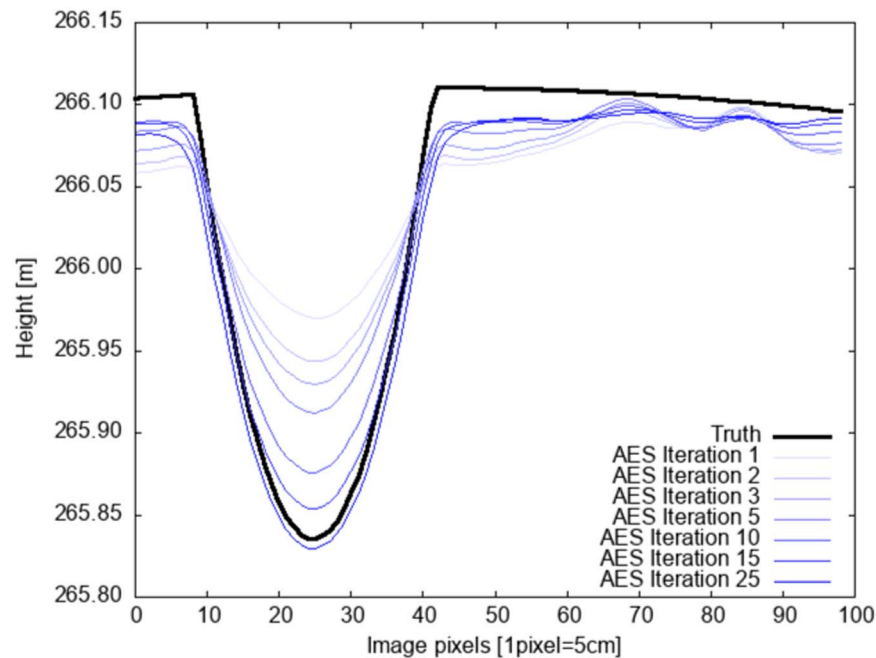


Figure 9. Vertical profile of a crater on an otherwise flat surface (from F2: Test 11). The black line is the truth terrain from which the source images were acquired. The colored lines show the terrain produced by SPC after each iteration step, indicating how numerous iterations are needed to converge on the truth topography.

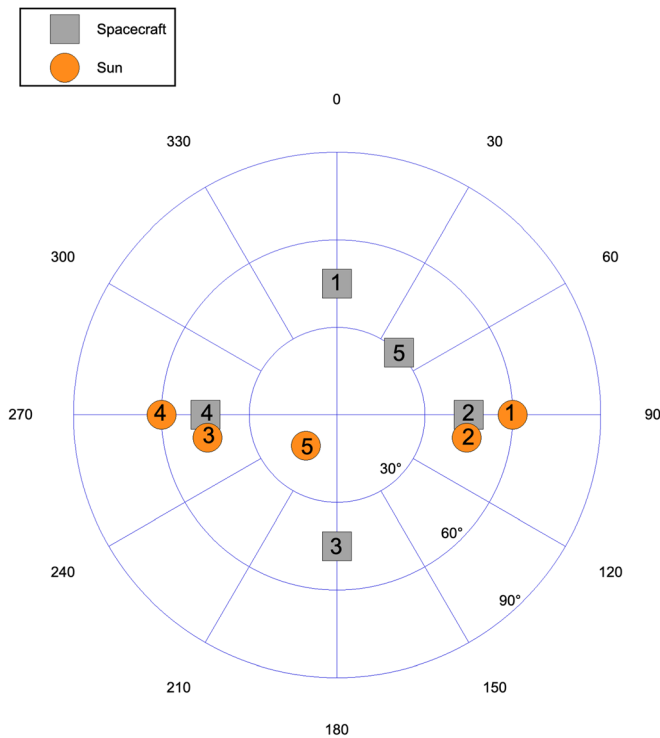


Figure 10. An example of a set of high-quality imagery for a piece of terrain. This example uses an asteroid with a near-zero obliquity, with the modeled location north of the equator by 20° . Topographic image stations are numbered 1–4, and the albedo image station is numbered 5.

Second, we evaluate how SPC processes the image data to identify two key components that must be considered: topographic expression and albedo expression. We use this conception/construct to evaluate which component each image provides to the data. While all images contain both elements, different images will better meet each of SPC's needs for slope and albedo.

By conducting tests predicated on this conception, we show the relative importance of each component to the solution and, as such, create a template for an optimum imagery observation plan. We show that five well-positioned images provide an exceptional solution, with additional images only providing limited improvement. Four of these images focus on extracting the topography and are positioned 90° in azimuth from one another. The fifth image supports the calculation of the surface albedo, meaning as low of an incidence angle as possible without introducing an opposition surge. Other aspects of the test allow us to identify that having a range of images in which the Sun is positioned on both sides of the topography (morning versus afternoon) is essential.

Testing also provides a foundation for understanding how many iterations of the AES cycle a good solution needs. We show that the solution approaches an asymptote such that additional processing is no longer effective. Ten iterations of the AES cycle balance the quality of the DTM to the processing time. The testing also showed that many different data sets could generate a high-quality DTM. If the mission cannot get the optimal suite of images, additional nonoptimal images and additional processing time can often mitigate the missing conditions.

Finally, this testing shows that the normalized cross-correlation technique provides a robust evaluation tool for evaluating the quality of DTMs. Since truth topography does not exist, this tool is exceptionally beneficial for real-world missions because it is based only on available data (i.e., images).

Ultimately, the test results were used to define the observation geometries that became part of the as-flown spacecraft flight profile (Lauretta et al. 2021). Specific recommendations for observation design that facilitates SPC shape modeling, based on the flight plan implemented by OSIRIS-REx, are provided in the [Appendix](#).

Acknowledgments

This material is based on work supported by NASA under contract NNM10AA11C issued through the New Frontiers Program and by the Canadian Space Agency. We are grateful to the entire OSIRIS-REx Team for making the encounter with Bennu possible. All data from the OSIRIS-REx mission are available via the Planetary Data System at <https://sbn.psi.edu/pds/resource/orex/>. These testing models are all archived in the Planetary Data System (Palmer & Gaskell 2024). The license of the PSI SPC software suite is available from the Planetary Science Institute at <https://www.psi.edu>. Please contact Eric Palmer (epalmer@psi.edu) for information on access, training, and the fee schedule. I'd like to thank two anonymous reviewers for contributions that greatly improved the quality and scope of this work.

Appendix

A.1. Topographic and Albedo Image Observing Requirements

The requirements for a high-quality image set are five images that meet the following:

Topographic Images—Four observation stations that focus on topographic expression, i.e., moderate emission angles

1. Emission vector geometry
 - (a) The emission angles' azimuth should be roughly N, S, E, W ($\pm 30^\circ$). It is desired that the differences in azimuth angles are $>80^\circ$ different, but they are required to be $>45^\circ$ different. The entire system can be rotated, but the difference between the viewing geometries should cover each side.
 - (b) Zenith angle: the spacecraft's view should be off-nadir with the desired zenith angle for four images between 20° and 45° , but at least three sets should be between 20° and 45° .
2. Solar illumination geometry (described by local hour)
 - (a) Required all $<16:00$ hr local ($i < 60^\circ$ at equator)
 - (b) Required: one set east:10:00 (desired within 1 hr, required within 2 hr)
 - (c) Required: one set west:14:00 (desired within 1 hr, required within 2 hr)
 - (d) Required: three sets must have local time different by one full hour

Albedo Image—One observing station is required that has a strong albedo signature.

1. Emission vector geometry
 - (a) Azimuth angle: roughly N or S
 - (b) Zenith angle: the desired zenith angle is between 0° and 15° , but it is required to be between 0° to 20°
2. Incidence vector geometry
 - (a) Local hour is required to be within 30 minutes of noon
 - (b) Azimuth angle can be any, with a desired variation between observation of $>90^\circ$

Overall Image Resolution

This discussion is designed to establish how many images with what pixel size are needed to build the DTM. Three general factors must be considered:

1. The images must be high enough resolution to build the model. There must be some images whose pixel size is

the same as or higher than the desired GSD. Some images can be a factor of two lower (worse resolution).

2. The images must be low enough resolution to register images to the existing DTM. The resolution must be at most five times better than the existing DTM. Every image must be aligned to the DTM. If the gap in resolution between the DTM and the image's pixel size is too large, then it cannot be registered (i.e., it cannot be located on the DTM). Thus, only two image stations must meet this requirement.
3. The image width cannot be less than twice the width of a maplet. If it is, then there is insufficient image fill onto the maplet (i.e., only a small section of the maplet would get data).

These constraints can be translated into the following measurable criteria. GSD is the GSD of the DTM that is desired. These resolution (image pixel size) requirements take advantage of SPC's ability to extrapolate data while still forming a high-quality DTM.

1. Resolution Criterion 1: Two image stations
 - (a) Image pixel size minimum: $\text{GSD}/5$
 - (b) Image pixel size maximum: GSD
2. Resolution Criterion 2: Three image stations
 - (a) Image pixel size minimum: $\text{GSD}/5$
 - (b) Image pixel size maximum: $\text{GSD} * 2$
3. Resolution Criterion 3: Two image stations
 - (a) Image pixel size minimum: Prior DTM's $\text{GSD}/5$.

ORCID iDs

Eric E. Palmer  <https://orcid.org/0000-0001-6755-8736>
 John R. Weirich  <https://orcid.org/0000-0002-2830-1708>
 Robert W. Gaskell  <https://orcid.org/0000-0002-2293-7879>
 Tanner Campbell  <https://orcid.org/0000-0001-9418-1663>
 Olivier S. Barnouin  <https://orcid.org/0000-0002-3578-7750>
 Michael G. Daly  <https://orcid.org/0000-0002-3733-2530>
 Kenneth Getzandanner  <https://orcid.org/0000-0002-7253-5931>
 John N. Kidd, Jr.  <https://orcid.org/0000-0002-4010-3050>
 Coralie D. Adam  <https://orcid.org/0000-0002-5795-2446>
 Dante S. Laurretta  <https://orcid.org/0000-0002-2597-5950>

References

- Adam, C., McCarthy, L. K., Leonard, J. M., et al. 2023, *PSJ*, 4, 167
 Al Asad, M. M., Philpott, L. C., Johnson, C. L., et al. 2021, *PSJ*, 2, 82
 Barnouin, O. S., Daly, M. G., Palmer, E. E., et al. 2020, *P&SS*, 180, 104764
 Craft, K. L., Barnouin, O. S., Gaskell, R., et al. 2020, *P&SS*, 193, 105077
 Daly, R. T., Ernst, C. M., Barnouin, O. S., et al. 2022, *PSJ*, 3, 207
 Ernst, C. M., Daly, R. T., Gaskell, R. W., et al. 2023, *EP&S*, 75, 103
 Gaskell, R. W., Barnouin, O. S., Daly, M. G., et al. 2023, *PSJ*, 4, 63
 Gaskell, R. W., Barnouin-Jha, O. S., Scheeres, D. J., et al. 2008, *M&PS*, 43, 1049
 Kirk, R. 1987, PhD thesis, Caltech
 Laurretta, D. S., Adam, C. D., Allen, A. J., et al. 2022, *Sci*, 377, 285
 Laurretta, D. S., Balram-Knutson, S. S., Beshore, E., et al. 2017, *SSRv*, 212, 925
 Laurretta, D. S., Enos, H. L., Polit, A. T., Roper, H. L., & Wolner, C. W. 2021, in Sample Return Missions: The Last Frontier of Solar System Exploration, ed. A. Longobardo (Amsterdam: Elsevier), 163
 Lewis, J. P. 1995, in Fast Normalized Cross-Correlation, Proc. of Vision Interface, 120
 Mario, C. E., Miller, C. J., Norman, C. D., et al. 2022, *PSJ*, 3, 104
 Nolan, M. C., Magri, C., Howell, E. S., et al. 2013, *Icar*, 226, 629
 Norman, C. D., Miller, C. J., Olds, R. D., et al. 2022, *PSJ*, 3, 101

- Olds, R., May, A., Mario, C., et al. 2015, in *Advances in the Astronautical Sciences Guidance, Navigation and Control 2015*, Vol. 154 (San Diego, CA: Univelt), <http://www.univelt.com/book=4963>
- Olds, R. D., Miller, C. J., Norman, C. D., et al. 2022, *PSJ*, **3**, 100
- Palmer, E. E., Weirich, J., Daly, M., et al. 2024, *PSJ*, **5**, 46
- Palmer, E. E., Gaskell, R., Daly, M. G., et al. 2022, *PSJ*, **3**, 102
- Palmer, E. E., & Gaskell, R. 2024, OSIRIS-REx Simulated Test Models V1.0, NASA Planetary Data System, <https://doi.org/10.26033/v0s3-z379>
- Palmer, E. E., Head, J. N., Gaskell, R. W., Sykes, M. V., & McComas, B. 2016, *E&SS*, **3**, 488
- Prockter, L., Murchie, S., Cheng, A., et al. 2002, *AcAau*, **51**, 491
- Weirich, J., Palmer, E. E., Daly, M. G., et al. 2022, *PSJ*, **3**, 103
- Yoshikawa, M., Kawaguchi, J., Fujiwara, A., & Tsuchiyama, A. 2021, in *Sample Return Missions: The Last Frontier of Solar System Exploration*, ed. A. Longobardo (Amsterdam: Elsevier), 123